Contacts: explanation and precise definitions

A companion to `two-proofs-reduced-inputs-v3.pdf`

Elementary meaning

The article fixes four distinct rational numbers and the symbol ∞. It calls these five values *contacts*. For a rational number $s = u/v$ in lowest terms, contact with 1 is measured by the prime factors of $u - v$: for example, 126 has contact depth 3 with 1 at 5, because $126 - 1 = 5^3$. For a fixed value $b = a/d$ in lowest terms, replace $u - v$ by $du - av$. For ∞, use the denominator v. The article also calls these expressions "contacts"; we distinguish the fixed values from the evaluated expressions below.

1. Definitions in terms of prime divisibility

For a prime q and a nonzero integer n, let $v_q(n)$ be the largest integer $e \geq 0$ such that $q^e \mid n$. For a nonzero rational $r = n/m$, put $v_q(r) = v_q(n) - v_q(m)$; this is independent of the representation. We call $v_q(r)$ the *valuation at q*. Set $v_q(0) = +\infty$. We use $v_q(rs) = v_q(r) + v_q(s)$ and, when $v_q(r) \neq v_q(s)$, $v_q(r + s) = \min\{v_q(r), v_q(s)\}$. Both identities follow by factoring out powers of q.

Write $\mathbb{P}^1(\mathbb{Q}) = \mathbb{Q} \cup \{\infty\}$. A pair $(u, v) \in \mathbb{Q}^2 \setminus \{(0, 0)\}$ represents u/v if $v \neq 0$, and represents ∞ if $v = 0$; the notation $(u : v)$ identifies pairs differing by a nonzero rational factor. A pair of integers is *primitive* if its coordinates have greatest common divisor 1. Each finite point has a unique primitive representation (u, v) with $v > 0$; for ∞ use $(1, 0)$.

Fix $C = \{\infty, b_1, b_2, b_3, b_4\}$ with distinct $b_i \in \mathbb{Q}$. Write $b_i = a_i/d_i$ with $\gcd(a_i, d_i) = 1$ and $d_i > 0$, and define integer linear polynomials

$$L_{b_i}(U, V) = d_i U - a_i V, \qquad L_\infty(U, V) = V.$$

For $s \in \mathbb{P}^1(\mathbb{Q}) \setminus C$, take its primitive pair (u, v). Its *contact depth at $b \in C$ and the prime q* is

$$\delta_q(s, b) = v_q(L_b(u, v)). \tag{1}$$

All these expressions are nonzero because $s \neq b$. Thus every depth is a finite nonnegative integer. We say there is *contact at q* when this depth is positive. In particular,

$$\delta_q(u/v, a_i/d_i) = v_q(d_i u - a_i v), \qquad \delta_q(u/v, \infty) = v_q(v).$$

2. What congruence means here

For primitive pairs (u, v) and (a, d), depth at least $k \geq 1$ means exactly $q^k \mid du - av$ (also for $(a, d) = (1, 0)$, which represents ∞). Equivalently, there is an integer λ with $q \nmid \lambda$ such that

$$u \equiv \lambda a \pmod{q^k}, \qquad v \equiv \lambda d \pmod{q^k}. \tag{2}$$

To prove this, if $q \nmid a$, choose $\lambda \equiv ua^{-1} \pmod{q^k}$. Divisibility of $du - av$ gives the second congruence. If $q \mid \lambda$, both u, v would be divisible by q, a contradiction. If $q \mid a$, then $q \nmid d$, and use $\lambda \equiv vd^{-1}$ instead. Conversely, (2) immediately gives $q^k \mid du - av$. The inverses exist because the inverted integers are coprime to q^k.

If $q \nmid vd$, this is ordinary congruence of fractions, computed by inverting their denominators modulo q^k; here $\delta_q(s, b) = v_q(s - b)$. Definition (1) also covers denominators divisible by q. At ∞ it says simply $q^k \mid v$.

3. Arbitrary rational coordinates

If (u, v) is any rational pair representing s, its relation to the primitive pair (u_0, v_0) is $(u, v) = \lambda(u_0, v_0)$ for $\lambda \in \mathbb{Q} \setminus \{0\}$. Hence

$$\delta_q(s, b) = v_q(L_b(u, v)) - \min\{v_q(u), v_q(v)\}. \tag{3}$$

Indeed, $L_b(u, v) = \lambda L_b(u_0, v_0)$ and the displayed minimum equals $v_q(\lambda)$. This proves the coordinate independence asserted in the article.

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4. The product of primes used in the article

For a finite set of primes S and $s \notin C$, define

$$M_S(s) = \prod_{\substack{q \notin S \text{ prime} \\ \delta_q(s,b) \text{ odd for some } b \in C}} q. \tag{4}$$

This is a finite positive integer with no repeated prime factor: each contributing prime divides one of the five nonzero integers $L_b(u,v)$ in primitive coordinates. An empty product is 1. "Odd" means a positive odd depth: depth 2 contributes nothing and depth 3 contributes one factor q. The article calls this the *contact radical*; equation (4) is its full definition.

One must test the five depths separately. They can be combined into the single exponent $v_q(\prod_{b\in C} L_b(u,v))$ if q divides none of the integers d_i or $a_i d_j - a_j d_i$ $(i \neq j)$. Indeed, if $q \mid v$, then $q \nmid u$ and none of the $d_i u - a_i v$ is divisible by q. If $q \nmid v$ and two finite expressions were divisible by q, then so would their combination $d_j L_{b_i}(u,v) - d_i L_{b_j}(u,v) = (a_j d_i - a_i d_j)v$, a contradiction. Thus at most one depth is positive. The listed integers are nonzero, so only finitely many primes are excluded. Without it, the parity of a sum can conceal two odd depths.

5. Raw contact expressions and the needed lemma

For arbitrary rational (u,v), a *raw contact expression* is the number $L_b(u,v)$. Its valuation can be negative and its parity can change when (u,v) is rescaled. Equation (3) removes precisely this dependence on the choice of coordinates.

Lemma (the article's Lemma B.5). Suppose $s = (u : v) \notin C$ and $q \nmid d_i$ for every i. If some $\delta_q(s,b)$ is odd, then $v_q(L_{b'}(u,v))$ is odd for some $b' \in C$.

Proof. Put $t = \min\{v_q(u), v_q(v)\} \in \mathbb{Z}$. Some raw expression has valuation exactly t. If $v_q(v) = t$, use $L_\infty = v$. Otherwise $v_q(u) = t < v_q(v)$, so $v_q(d_i u) = t$ and $v_q(a_i v) > t$; hence $v_q(L_{b_i}(u,v)) = t$. If t is odd, this proves the claim. If t is even, use (3) for a contact of odd depth. □

Here an odd raw valuation may be negative. The lemma asserts only this parity implication; it does not assert its converse. For example, let $C = \{\infty, 0, 1, 2, 3\}$, $s = 4$, and $q = 5$. All five depths are zero, whereas the coordinates $(u,v) = (20,5)$ give raw values $5, 20, 15, 10, 5$, all of valuation 1 at 5. For the remainder, assume S contains every prime dividing some d_i, as in the article. The lemma then applies to every prime counted in $M_S(s)$.

6. Why there are also "thirteen contacts"

For three slopes $s_i = u_i/v \notin C$ with the same nonzero denominator v, the raw expressions are

$$v, \qquad L_b(u_i, v) \quad (1 \le i \le 3,\ b \in C \setminus \{\infty\}).$$

These are $1 + 3 \cdot 4 = 13$ indexed expressions, even if some values happen to coincide. The five distinguished points remain unchanged. By the lemma, each prime dividing any $M_S(s_i)$ gives an odd valuation of at least one of these thirteen expressions.

7. The elementary inference used in the height argument

Let $\mathcal{D}$ be the set of primes outside S at which at least one of these thirteen expressions has odd valuation. Suppose $D \neq 0$ is an integer and $q^K \mid D$ for every $q \in \mathcal{D}$, where $K \ge 1$ is an integer. The lemma and unique prime factorization give $M_S(s_i)^K \mid D$ for each i.

For primitive (u,v), define the logarithmic height by $h((u : v)) = \log \max\{|u|, |v|\}$, with the natural logarithm. The main article supplies, under its arithmetic assumptions, a constant $H_d \ge 1$ such that $h(s) \le H_d M_S(s)^{24}$. Consequently, with $K = 24$,

$$h(s_i) \le H_d M_S(s_i)^{24} \le H_d |D|.$$

These definitions and elementary lemmas require no unproved arithmetic input. The height bound and the construction enforcing $q^{24} \mid D$ are separate parts of the main article (§§1–3 and Appendix B).

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